

MONMOUTH JET CENTER, NJ, USA

WMO: 724084

Lat: 40.183N

Lon: 74.117W

Elev: 159

StdP: 14.61

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 54760

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.8	(c) 15.8	(d) -8.5	(e) 3.5	(f) 19.4	(g) -3.6	(h) 4.6	(i) 18.3	(j) 28.7	(k) 31.4	(l) 26.2	(m) 30.9	(n) 10.9	(o) 290	(p) 0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.4	(c) 91.1	(d) 74.3	(e) 88.5	(f) 73.1	(g) 85.7	(h) 72.2	(i) 77.3	(j) 86.9	(k) 75.8	(l) 84.3	(m) 74.4	(n) 82.0	(o) 10.0	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	130.3	81.7	73.0	123.3	79.8	72.1	119.4	79.2	40.7	87.3	39.3	84.4	37.8	81.2	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a) 24.8	(b) 21.3	(c) 18.9	(e) 5.1	(f) 96.1	(g) 6.1	(h) 2.4	(i) 0.7	(j) 97.8	(k) -2.8	(l) 99.2	(m) -6.2	(n) 100.6	(o) -10.6	(p) 102.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	32.4	34.7	41.0	51.5	60.5	70.2	75.7	73.7	67.4	55.9	46.4	37.9	
	DBStd	16.93	10.54	9.52	9.07	8.50	7.73	6.58	5.26	5.15	6.16	7.68	8.49	9.17	
	HDD50	1961	553	436	306	76	3	0	0	0	28	167	392		
	HDD65	4982	1011	849	745	418	189	27	1	2	43	297	559	841	
	CDD50	3434	7	7	27	120	328	605	796	734	523	212	59	16	
	CDD65	975	0	0	1	12	48	181	331	270	116	16	0	0	
	CDH74	7467	0	1	12	130	439	1387	2825	1954	640	77	2	0	
	CDH80	2440	0	0	2	39	133	442	1058	600	154	12	0	0	
Wind	WSAvg	8.6	9.9	10.1	10.5	10.0	8.1	7.6	6.9	6.6	7.4	8.1	8.8	9.3	
Precipitation	PrecAvg	47.6	3.9	2.7	4.5	3.7	3.6	4.0	4.4	4.6	4.3	4.0	3.4	4.3	
	PrecMax	57.7	7.9	5.6	9.9	9.7	8.0	7.8	7.0	10.0	8.3	12.4	6.7	8.4	
	PrecMin	35.4	1.5	0.5	0.9	1.3	0.9	1.2	1.0	1.3	1.1	0.7	0.8	1.1	
	PrecStd	5.2	1.5	1.3	2.0	1.9	1.6	1.8	1.6	2.0	2.1	2.7	1.6	1.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.3	66.3	72.7	84.7	89.6	92.5	96.9	93.3	89.7	81.3	71.7	66.0	
		MCWB	55.6	55.5	58.0	64.7	69.9	74.2	76.2	77.1	73.9	67.0	61.8	58.9	
	2%	DB	57.3	58.6	64.3	77.3	83.7	89.6	91.9	89.8	84.0	75.1	66.0	61.0	
		MCWB	51.0	51.1	53.3	61.3	67.4	73.1	75.0	74.3	71.2	65.8	58.2	55.2	
	5%	DB	52.9	54.0	60.5	70.4	79.2	84.4	89.8	86.1	81.0	72.0	63.1	55.5	
		MCWB	47.7	47.4	50.5	57.7	64.9	70.9	74.0	73.2	68.8	63.9	55.9	50.6	
	10%	DB	46.7	47.9	54.8	64.8	73.5	81.6	86.0	82.5	77.3	68.1	59.7	52.2	
		MCWB	41.6	42.2	47.1	54.4	61.5	69.2	73.2	71.1	67.7	61.7	54.0	47.3	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.3	59.5	60.6	66.7	73.0	76.6	80.0	79.3	76.9	72.2	63.9	60.4	
		MCDB	62.3	64.9	70.7	78.8	83.3	87.7	91.6	89.4	85.6	75.6	69.2	63.9	
	2%	WB	53.3	52.7	56.1	63.5	70.0	74.9	77.6	77.1	74.3	68.6	60.3	56.7	
		MCDB	57.0	57.1	62.2	74.2	79.8	85.1	87.7	85.8	79.6	73.7	64.2	60.4	
	5%	WB	48.0	47.0	52.1	60.0	67.2	73.0	76.2	75.3	72.2	65.5	57.8	52.0	
		MCDB	51.8	52.3	58.4	68.8	76.1	82.1	85.1	82.7	77.4	69.8	61.9	55.9	
	10%	WB	42.7	42.5	47.4	56.8	64.1	71.2	74.7	73.6	70.1	62.4	54.6	47.4	
		MCDB	46.3	47.6	54.8	64.1	71.8	79.4	82.8	79.8	75.5	66.9	58.8	51.4	
Mean Daily Temperature Range			MDBR	14.0	15.0	15.5	16.9	16.8	16.5	16.4	15.6	15.9	16.5	15.5	14.0
	5% DB	MCDBR	19.7	21.8	23.1	24.2	23.7	20.8	20.2	18.8	19.0	19.8	18.8	17.8	
		MCWBR	17.2	16.9	15.6	13.0	11.9	9.6	7.9	8.4	9.9	12.1	14.0	16.2	
		MCDBR	19.3	20.4	20.5	21.1	20.5	18.5	17.2	16.1	15.4	15.9	16.0	17.4	
	5% WB	MCWBR	18.1	17.4	16.3	13.3	11.2	9.8	7.8	8.0	9.3	11.8	14.2	17.3	
Clear-Sky Solar Irradiance	taub		0.315	0.324	0.357	0.381	0.425	0.480	0.481	0.472	0.397	0.361	0.331	0.313	
	taud		2.482	2.460	2.412	2.373	2.282	2.166	2.175	2.193	2.374	2.480	2.519	2.534	
	Ebn at Noon		270	284	284	284	272	256	254	253	267	266	261	262	
	Edh at Noon		24	29	34	37	42	47	46	44	35	28	23	22	
All-Sky Solar Radiation	RadAvg		607	897	1210	1555	1725	1897	1926	1684	1394	983	692	528	
	RadStd		54	93	101	141	189	170	94	113	132	96	62	50	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (61 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+214	+124

Nomenclature: See separate page